

Activate Your Core Workbook

OVERVIEW

[Overview icon]

WELCOME TO THE ACTIVATE YOUR CORE WORKBOOK FOR GEOLOGY 1403!

This workbook serves as an active learning space where you can deeply engage in your learning experience in this core academic course.

Embedded with guiding questions and activities as well as supportive tools, this workbook is designed to encourage profound introspection into your learning journey and mastery of course material. Ultimately, it will enable you to not only reflect on *what* you learn and *how* you learn, but most importantly *why* learning this content is critical to your growth as a learner and a teacher.

Ultimately, this Core Workbook will help you:

- Identify your learning achievements
- Identify and reflect on your learning challenges
- Determine the learning behaviors, strategies, and skills that support your learning success
- Connect what you are learning to key course learning outcomes and EC-6 competency needs
- Create a powerful review tool as you prepare for the Texas Core Subjects EC-6 (391) educator certification exams

The *Activate Your Core* Workbook has two sections.

SECTION 1: ACTIVATE YOUR CORE

This first section of your workbook:

- Is divided into eight chapters, one for each module of the course
- Will help support your **acquisition** and **retention** of course-specific **knowledge, skills, and strategies**
- Includes questions and prompts to encourage you to think critically about your learning as you engage with the learning materials, activities, and assignments throughout the course
- Will enable you to create a learning record you can refer to later in preparation for the EC-6 subject area exams.

SECTION 2: MY CORE GALLERY

The second section of your workbook:

- Will serve as the showcase of core pieces of work that you completed or created during the course
- Will capture your ability to engage with **what** you learned
- Will demonstrate **how** you learned
- Will provide a general sense of your overall learning in this course and **why** it matters

Activate Your Core Workbook

GEOL 1403: Physical Geology

Physical Geology

Section I: Activate Your Core

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Section I: Activate Your Core

MODULE 0: COURSE INTRODUCTION

BEFORE YOU START THE COURSE

[Metacognition icon]

Welcome to our reflective activity! Before diving into our course, take a moment to reflect on where you stand. Think about your current knowledge, competency, and skill level in geology.

This self-assessment will help you gauge your starting point and set realistic expectations for your journey ahead. Additionally, consider any aspects of the course that might be causing you a bit of concern or anxiety. Identifying these areas early on will allow for a smoother learning experience.

Respond to the following questions in at most 100 words.

1. How would you describe your knowledge level, competency level, and/or skill level in geology before taking this course? The table below will help you organize your thoughts

KNOWLEDGE (for example: <i>definitions, concepts, key terms, theorems, theories, facts, themes, topics, events, causes, effects, etc.</i>)	SKILLS (for example: <i>computation, note-taking, problem-solving, critical thinking, research, critical reading, communication, time management, etc.</i>)	STRATEGIES (for example: <i>making an outline or timeline, using a graphic organizer, reviewing reading notes and video lecture notes, asking questions, completing exam wrappers, practicing, self-explanation, etc.</i>)

2. What components of the course are you a little concerned or anxious about? Why?

MODULE 1 | THE NATURE OF GEOLOGY

MODULE 1: MODULE OVERVIEW

MY PERSONAL LEARNING GOALS FOR MODULE 1:

- Preview **Module 1**.
- Review the **Module Overview page**, where you can review the Learning Activities list for the module.
- Scan each page of the module.
- Complete the Metacognitive Moment questions listed below.

[ICON] METACOGNITIVE MOMENT

- BASED ON THE MODULE OVERVIEW, WHAT DO YOU ALREADY KNOW ABOUT THIS TOPIC?
- HOW MUCH TIME WILL YOU NEED TO COMPLETE THIS MODULE?
- HOW CAN I MOTIVATE MYSELF TO DO THE MODULE ACTIVITIES AND ASSIGNMENTS TO THE BEST OF MY ABILITY?

Record your desired learning goals for this module in the space below.

MY PERSONAL LEARNING GOALS FOR THIS WEEK ARE:

Record the days/times you plan to work on the module and schedule this time in your calendar to fulfill your learning path for Module 1 of the course.

MY LEARNING PATH FOR MODULE 1:

MODULE 1: LEARNING MATERIALS

MODULE 1 READING NOTES AND RESPONSES TO KEY QUESTIONS:

MODULE 1 READING:

Chapter 1: An Introduction to Geology

- Section 1.5: Geology: Origin and Early Evolution of Earth
- Section 1.6: Earth's Internal Structure

Read pp. 21–23

Review the Chapter 1 PowerPoint: [01_earth13_lecture_ppt](#)

Key Question(s):

- How do the different stages of solar system formation relate to the current arrangement and characteristics of planets we observe today?
- In what ways does Earth's internal structure influence surface phenomena such as plate tectonics, volcanism, and earthquakes?

MODULE 1 READING:

Chapter 1: An Introduction to Geology

- Section 1.3: The Nature of Scientific Inquiry

Read pp. 11–12

Review the Chapter 1 PowerPoint: [01_earth13_lecture_ppt](#)

Key Question(s):

- How does the process of developing and testing hypotheses in geology differ from or resemble the scientific method used in other natural sciences?
- What unique challenges might geologists face in this process?

MODULE 1 READING:

Chapter 1: An Introduction to Geology

- Section 1.4: Earth as a System

Read pp. 13–16

Key Question(s):

- How might a change in one of Earth's spheres potentially affect the others? Can you think of a specific example to illustrate this interconnectedness?
- In what ways do humans interact with and influence each of Earth's four major spheres? Consider both positive and negative impacts.

MODULE 1: LEARNING MATERIALS HIGHLIGHTS

Record the most meaningful, memorable, or insightful quote(s) from this week's readings and/or video lectures and why you feel that way.

MY QUOTES FOR MODULE 1:

MODULE 1: ACTIVITIES AND ASSIGNMENTS

MODULE 1: INTRODUCTION TO GEOLOGY

EC-6 CONNECTION

This learning activity connects to the following competency(ies) and sub-skill(s) assessed in the **EC-6 (391) Texas Educator Certification Examination for Geology**.

Competency 004—(Concepts and Processes): The teacher knows and understands the unifying concepts and processes that are common to all sciences.

The beginning teacher:

- A. Understands how a unifying, explanatory framework across the science disciplines is provided by the concepts and processes of systems, order, and organization; evidence, models, and explanation; change, constancy, and measurements; and form and function.

Competency 015—(Structure and Function of Earth Systems): The teacher understands the structure and function of Earth systems.

The beginning teacher:

- A. Understands the structure of Earth and analyzes constructive and destructive processes (including plate tectonics, weathering, and erosion) that produce geologic change, including how these processes have affected Earth history.

MODULE 1: VOCABULARY ACTIVITY

Instructions: Below is a word bank and a table with two columns. The first column contains definitions, and the second column contains vocabulary words. Your task is to match each definition with the correct vocabulary word by writing the vocabulary word in the appropriate space.

Word Bank

- Geology
- Physical Geology

- Historical Geology
- Earth Systems
- Hydrosphere
- Atmosphere
- Biosphere
- Geosphere
- Uniformitarianism
- Hypothesis
- Theory

Table

Definition	Vocabulary Word
The study of the Earth, its materials, and the processes acting upon them.	
The branch of geology concerned with the physical features and processes of the Earth.	
The branch of geology that deals with the history of the Earth as recorded in rocks.	
The interconnected systems that make up the Earth, including the atmosphere, hydrosphere, biosphere, and geosphere.	
The water component of the Earth, including oceans, lakes, rivers, and groundwater.	
The layer of gasses surrounding the Earth, held by gravity.	
The regions of the Earth where living organisms exist.	
The solid part of the Earth, including the crust, mantle, and core.	
The principle that the same natural laws and processes that operate in the universe now have always operated in the universe and apply everywhere in the universe.	
A proposed explanation for a phenomenon, which can be tested through experimentation and observation.	

A well-substantiated explanation of some aspect of the natural world, based on a body of evidence.	
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Reflect and respond:

1. Were there any words or definitions that you found particularly challenging? If so, which ones and why?
2. How did you overcome these challenges?

MODULE 1: STENO'S PRINCIPLES ASSIGNMENT EC-6 CONNECTION

This learning activity connects to the following competency(ies) and sub-skill(s) assessed in the **EC-6 (391) Texas Educator Certification Examination for Geology** .

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MODULE 1: STENO'S PRINCIPALS' INTERACTIVE CONNECTION

Introduction to Steno's Principles

Review: Nicolas Steno was a pioneering geologist whose principles are foundational to the study of stratigraphy. Here are the three main principles you need to know:

1. Principle of Superposition: In undisturbed layers of rock, the oldest layers are at the bottom, and the youngest layers are at the top.
2. Principle of Original Horizontality: Layers of rock are originally deposited horizontally.
3. Principle of Lateral Continuity: Layers initially extend laterally in all directions.

Virtual Game Instructions

Access the Game: Click on the link to [Layers of Time | AMNH](#) to access the virtual game platform.

Objective of the Game: Your goal is to arrange the given rock layers from oldest to newest using Steno's principles.

Playing the Game: Drag and drop the rock layers to place them in the correct chronological order. Use the principles of superposition, original horizontality, and lateral continuity to guide your decisions. Consider the Metacognitive Reflection questions below prior to the start of your game.

Metacognitive Reflection

After completing the game, take a few moments to reflect on your thought process. Use the following prompts to guide your reflection. Responses are not required.

Before Playing:

What prior knowledge did you have about Steno's principles?
What strategies did you plan to use to arrange the rock layers?

During the Game:

What challenges did you encounter while arranging the layers?
How did you overcome these challenges?

After Playing:

What did you learn about Steno's principles through this activity?
How did reflecting on your thought process help you understand the material better?

Reflection Entry

Write a brief reflection below addressing the following questions:

- How did Steno's principles help you determine the order of the rock layers?
- How did reflecting on your thought process enhance your understanding of stratigraphy?
- How did the gamification and visualization aspects of this activity aid in your learning retention and understanding of the concepts?

MODULE 1: MONITOR YOUR PROGRESS

Take a moment to monitor your progress in this module by addressing the questions below:

- Are you using the best strategies to learn this material? If not, what could you do differently?
- Are you going too fast through the module? Are you going too slowly? (One way to reflect on this question is to think about which areas you moved through quickly, and which you took more time on. How is the speed of your progress through the module affecting your learning and the quality of your assessments?)

-
- Are you understanding the information (or mastering the skills) in this module? If not, what can you do?

Citation: Rasmussen, C. L., & Stewart, P. W. (2018). Metacognitive prompts within an online course. *Journal of the International Society for Teaching Education*, 22(1), 19–29. <https://files.eric.ed.gov/fulltext/EJ1237532.pdf>

MODULE 1: QUIZ

MODULE 1: QUIZ WRAPPER

- How did you study for this quiz? What strategies did you use to prepare? Which strategies seemed most effective?
- What study practices did not work well? What should you not do, or do differently, next time?
- On which aspects of this quiz did you perform well? Why do you think you performed well on these aspects?
- Can you see patterns in your error and/or specific subjects? How can you address them in future preparation?

MODULE 1: WRAP-UP

THINK ABOUT YOUR LEARNING IN THIS MODULE BY ADDRESSING THE QUESTIONS BELOW:

- How well did you do on this module?
- Which strategies worked best for you for each activity/assignment in this module?
- What are your most significant learnings, or “Aha!” moments, from this module?
- What do these significant learnings, or “Aha!” moments, reveal to you or help you realize about principles of superposition?

-
- Before moving on, select the work piece that captures your most significant learning achievement in this module.
 - Document and archive this work piece in the [My Core Gallery](#) section of this workbook.

MODULE 2 | EARTH'S STRUCTURE AND FUNCTION

MODULE 2: MODULE OVERVIEW

MY PERSONAL LEARNING GOALS FOR MODULE 2:

- Preview **Module 2**.
- Review the **Module Overview page**, where you can review the Learning Activities list for the module.
- Scan each page of the module.
- Complete the Metacognitive Moment questions listed below.

[ICON] METACOGNITIVE MOMENT

- BASED ON THE MODULE OVERVIEW, WHAT DO YOU ALREADY KNOW ABOUT THIS TOPIC?
- HOW MUCH TIME WILL YOU NEED TO COMPLETE THIS MODULE?
- HOW CAN I MOTIVATE MYSELF TO DO THE MODULE ACTIVITIES AND ASSIGNMENTS TO THE BEST OF MY ABILITY?

Record your desired learning goals for this module in the space below.

MY PERSONAL LEARNING GOALS FOR THIS WEEK ARE:

Record the days/times you plan to work on the module and schedule this time in your calendar to fulfill your learning path for Module 2 of the course.

MY LEARNING PATH FOR MODULE 2:

MODULE 2: LEARNING MATERIALS

MODULE 2 READING NOTES AND RESPONSES TO KEY QUESTIONS:

MODULE 2 READING:

Chapter 2: Plate Tectonics: A Scientific Revolution Unfolds

Read pp. 36–69

Review the Chapter 2 PowerPoint: [02_earth13_lecture_ppt](#)

Key Question(s):

- How did the accumulation of evidence over time lead to the shift from the continental drift hypothesis to the more comprehensive plate tectonics theory?
- In what ways do the interactions at different types of plate boundaries contribute to the overall dynamics of Earth's surface and interior?

MODULE 2 READING:

Chapter 5: Volcanoes and Volcanic Hazards

Read pp. 134–171

Review the Chapter 5 PowerPoint: [05_EARTH13_Lecture_PPT](#)

Key Question(s):

- How do the different types of volcanoes and their eruption styles reflect the underlying geological processes and magma composition?
- In what ways does our understanding of volcanic processes and monitoring techniques contribute to hazard mitigation and public safety in volcanically active regions?

MODULE 2 READING:

Chapter 11: Earthquakes and Earthquake Hazards

Read pp. 310–339

Review the Chapter 11 PowerPoint: [11_earth13_lecture_ppt](#)

Key Question(s):

- How does the relationship between plate tectonics and earthquake distribution inform our understanding of seismic risk in different regions?
- In what ways do the limitations of current earthquake prediction methods impact disaster preparedness and risk management strategies?

MODULE 2 READING:

Chapter 12: Earth's Interior

Read pp. 340–361

Review the Chapter 12 PowerPoint: [12_earth13_lecture_ppt](#)

Key Question(s):

- How do the different types of volcanoes and their eruption styles reflect the underlying geological processes and magma composition?
- In what ways does our understanding of volcanic processes and monitoring techniques contribute to hazard mitigation and public safety in volcanically active regions?

MODULE 2: LEARNING MATERIALS HIGHLIGHTS

Record the most meaningful, memorable, or insightful quote(s) from this week's readings and/or video lectures and why you feel that way.

MY QUOTES FOR MODULE 2:

MODULE 2: ACTIVITIES AND ASSIGNMENTS

MODULE 2: UNDERSTANDING PLATE TECTONICS ASSIGNMENT

EC–6 CONNECTION

This learning activity connects to the following competency(ies) and sub-skill(s) assessed in the **EC–6 (391) Texas Educator Certification Examination for Geology**.

Competency 004—(Concepts and Processes): The teacher knows and understands the unifying concepts and processes that are common to all sciences.

The beginning teacher:

- Understands how a unifying, explanatory framework across the science disciplines is provided by the concepts and processes of systems, order, and organization; evidence, models, and explanation; change, constancy, and measurements; and form and function.
- Demonstrates an understanding of how patterns in observations and data can be used to make explanations and predictions.
- Analyzes interactions and interrelationships between systems and subsystems.

H. Understands how models are used to represent the natural world and how to evaluate the strengths and limitations of a variety of scientific models (e.g., physical, conceptual, mathematical).

Competency 007—(Forces and Motion): The teacher understands forces and motion and their relationships.

The beginning teacher:

D. Analyzes the relationship between force and motion in a variety of situations (e.g., simple machines, geologic processes).

Competency 009—(Energy and Interactions): The teacher understands energy and interactions between matter and energy.

The beginning teacher:

B. Understands the basic concepts of heat energy and related processes (e.g., melting, evaporation, boiling, condensation, conduction, convection, and radiation).

MODULE 2: PLATE TECTONICS

Learning Objective: Understand the basic principles of plate tectonics, and explain how convection currents in the mantle contribute to plate movement. Relate the behavior of a lava lamp to the movement of tectonic plates. Differentiate between convection, conduction, and radiation in the context of Earth's processes.



A lava lamp is a decorative lamp that contains a special kind of wax inside a liquid. When the lamp is turned on, a light bulb heats the wax, causing it to expand and rise through the liquid. As the wax reaches the top of the lamp, it cools, contracts, and sinks back down, creating a continuous cycle of rising and falling blobs of wax. This mesmerizing movement is the perfect example of a convection current, similar to the processes occurring in the Earth's mantle. For reference, you can view a lava lamp in action in the video [“Orange Lava Lamp: 8 Hours of Relaxation & Sleep Aid in 4K UHD.”](#)

Observation of the Lava Lamp:

- Watch the video of a lava lamp or observe a physical lava lamp if available. Pay attention to how the wax heats up, rises, cools down, and sinks back down.
- Describe what you observe. Include details about the movement of the wax and how it behaves as it heats and cools.

Connecting Lava Lamp Behavior to Plate Tectonics:

- How does the movement of the wax in the lava lamp simulate convection currents in the Earth's mantle?
- What causes the wax to rise and fall in the lava lamp, and how is this similar to the movement of material in the mantle?

- c. Explain how convection currents in the mantle drive the movement of tectonic plates. Use the lava lamp as a model to illustrate your explanation.
- d. Describe the types of plate boundaries and the geological features associated with each. How do convection currents affect these boundaries?
- e. How do convection, conduction, and radiation each contribute to the transfer of heat within the Earth's interior?

Application and Reflection: Reflect on how this model helps you understand the processes occurring beneath the Earth's surface. Answer the following metacognition questions in your Google Doc workbook:

- a. How did observing the lava lamp help you understand the concept of mantle convection and plate tectonics?
- b. What aspects of the lava lamp model were most helpful, and what aspects might be misleading or overly simplistic?

Micro Extension Activity: Research a specific tectonic plate boundary (e.g., the San Andreas Fault, the Mid-Atlantic Ridge, or the Himalayan mountain range) and describe how mantle convection influences the geological activity in that region. Summarize your findings in 4–6 sentences:

MODULE 2: UNDERSTANDING PLATE TECTONICS ASSIGNMENT

EC–6 CONNECTION

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- B. Demonstrates an understanding of how patterns in observations and data can be used to make explanations and predictions.
- C. Analyzes interactions and interrelationships between systems and subsystems.
- H. Understands how models are used to represent the natural world and how to evaluate the strengths and limitations of a variety of scientific models (e.g., physical, conceptual, mathematical).

Competency 007—(Forces and Motion): The teacher understands forces and motion and their relationships.

The beginning teacher:

- D. Analyzes the relationship between force and motion in a variety of situations (e.g., simple machines, geologic processes).

Competency 009—(Energy and Interactions): The teacher understands energy and interactions between matter and energy.

The beginning teacher:

- B. Understands the basic concepts of heat energy and related processes (e.g., melting, evaporation, boiling, condensation, conduction, convection, and radiation).

MODULE 2: CONTINENTAL DRIFT VS. PLATE TECTONICS

In the early 20th century, Alfred Wegener proposed a hypothesis that the continents were drifting. He suggested that Earth's continents were once part of a supercontinent called Pangaea that broke apart around 200 million years ago. He supported his hypothesis with evidence such as the fit of continental coastlines, similar rock formations, and identical fossil records found on different continents. Despite this, Wegener's ideas were initially rejected due to the lack of a convincing mechanism for continental movement. It wasn't until the 1960s, with the advancement of ocean floor exploration and new evidence such as seafloor spreading and the formation of mid-ocean ridges, that Wegener's ideas were validated and widely accepted, revolutionizing our understanding of Earth's dynamic surface. This led to the development of the Theory of Plate Tectonics.

Instructions

Using the information from your textbook (Chapter 2) and the provided text blurb, complete the following comparison chart. This chart will help you understand the differences between the pre-1960s understanding of continental drift and the post-1960s theory of plate tectonics.

Comparison Chart: Continental Drift vs. Plate Tectonics

Aspect	Pre-1960s Understanding (Continental Drift)	Post-1960s Understanding (Plate Tectonics)
Evidence 1		
Evidence 2		
Mechanism		
Scientific Acceptance		

Aspects to Consider:

1. **Evidence:** What types of evidence supported Wegener's hypothesis? What new evidence emerged in the 1960s?
2. **Mechanism:** What mechanisms did Wegener propose, and what mechanisms are now understood to drive plate movements?
3. **Scientific Acceptance:** How was Wegener's hypothesis received by the scientific community compared to the acceptance of plate tectonics?

MODULE 2: PLATE BOUNDARIES

Objective: Students will be able to identify and describe different types of plate boundaries, including divergent, convergent, and transform boundaries. They will illustrate these boundaries with diagrams, identify the tectonic plates involved, and explain the resulting landforms associated with each type of boundary.

Introduction: Plate boundaries are regions where two tectonic plates meet and interact, shaping the Earth's surface through various geological processes. There are three main types of plate boundaries: divergent boundaries, where plates move apart and new crust forms; convergent boundaries, where plates collide, leading to the formation of mountains, trenches, and volcanic activity; and transform boundaries, where plates slide past each other, causing earthquakes along faults.

Recent discoveries have significantly expanded our understanding of these dynamic regions. One notable finding is the identification of a global layer of partially molten rock beneath Earth's tectonic plates. Located within the asthenosphere, this layer was previously thought to exist in isolated patches, but recent studies have revealed its widespread presence. Interestingly, despite this extensive melt, it has minimal impact on mantle flow and plate movements, contrary to prior assumptions. (For additional information, see: [Scientists Detect Global Layer of Melt Hidden Below Earth's Tectonic Plates.](#))

Additionally, researchers have uncovered remnants of a previously unknown tectonic plate named Pontus, which existed around 150 million years ago. This ancient plate was identified through the study of mountain belts and rocks across regions from Japan to New Zealand, highlighting the complex and dynamic history of tectonic movements in the Earth's past. (For additional information, see: [Surprising Tectonic Discovery: Geologist Unexpectedly Finds Remnants of a Lost Mega-Plate.](#))

Adding to the intrigue, scientists have discovered that the Pacific plate is being torn apart by large undersea faults. This challenges previous notions about the robustness of oceanic plates and suggests that even these massive structures can be significantly damaged over time. Such findings are linked to seismic activity and volcanism, reshaping our understanding of these fundamental geological processes. (For additional information, see: [Geoscientists Find Pacific Plate Is Scored by Large Undersea Faults That Are Pulling It Apart.](#))

Instructions:

1. **Identify Plate Boundary Types:** Research and identify the three main types of plate boundaries: divergent, convergent, and transform.
2. **Describe Plate Movements:** For each type of boundary, describe the movements of the tectonic plates involved. Include details such as the direction of movement and any subduction processes.

3. **Find Relevant Images:** Use the internet to find images that accurately depict each type of plate boundary. Ensure the images are clear and correctly labeled with the type of boundary.
4. **Identify Resulting Landforms:** List the landforms that are typically created by each type of plate boundary. Provide examples of real-world locations where these landforms can be observed.

Types of Boundaries			
	Description of plates and their relationship to each other	Diagram or image of each, screenshot or search engine image	List the resulting landform/s
Divergent Boundary E-Text 2.6			
Convergent Boundary (Subduction) E-Text 2.5			
Convergent Boundary (Collisional) E-Text 14.1			
Transform Boundary E-Text 2.6			

MODULE 2: IDENTIFYING PLATE BOUNDARIES

Objective: Students will analyze descriptions of four tectonic locations to identify and explain the types of plate boundaries (divergent, convergent subduction, transform, and convergent collisional) present at each location.

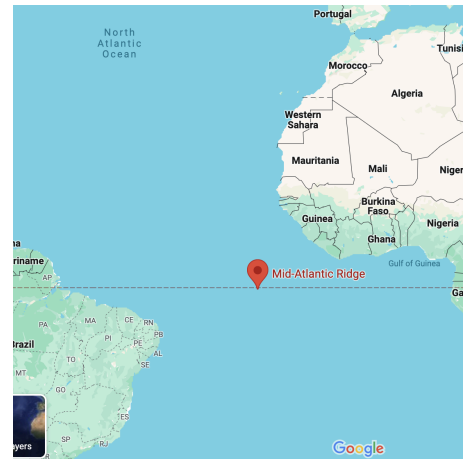
Instructions: Read the detailed paragraphs provided below for each of the locations. Based on the description, identify the type of plate boundary present at each location. Choose from the following boundary types:

- Divergent
- Convergent Subduction
- Transform
- Convergent Collisional

Write your answers in the provided spaces and explain your reasoning in 3–4 sentences for each location.

1. Mid-Atlantic Ridge

The Mid-Atlantic Ridge is an underwater mountain range that runs down the center of the Atlantic Ocean. It is one of the most prominent features of the ocean floor and stretches over 16,000 kilometers. This ridge is characterized by a continuous line of volcanic activity, where new oceanic crust is formed as magma rises from below the Earth's surface. This process causes the seafloor to spread apart, creating new oceanic crust. Interesting fact: In 2023, scientists recorded a significant earthquake of magnitude 5.9 along this ridge, highlighting the ongoing tectonic activity.



Boundary Type:

Reasoning:

2. Pacific Rim

The Pacific Rim, also known as the "Ring of Fire," is a region encircling the Pacific Ocean characterized by frequent earthquakes and numerous active volcanoes. This area is where oceanic plates are being forced down into the mantle beneath lighter continental plates, leading to intense seismic activity and volcanic eruptions. Notably, in 2011, the devastating Tōhoku earthquake and tsunami occurred along this boundary, highlighting the destructive potential of subduction zones. The area remains one of the most geologically active regions on Earth.



Boundary Type:

Reasoning:

this

3. Himalayas

The Himalayas, home to the world's highest peaks including Mount Everest, are located in Asia between the Indian subcontinent and the Tibetan Plateau. This mountain range is the result of the collision between the Indian Plate and the Eurasian Plate, which began around 50 million years ago and continues today. The immense pressure from this collision has caused the crust to buckle and rise, forming the towering mountains. In 2015, Nepal experienced a catastrophic earthquake with a magnitude of 7.8, demonstrating the ongoing tectonic activity in the region.

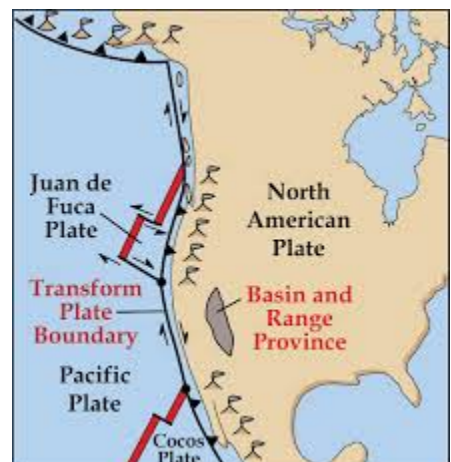


Boundary Type:

Reasoning:

4. Boundary between the Juan de Fuca Plate and the North American Plate

The boundary between the Juan de Fuca Plate and the North American Plate is located off the coast of the Pacific Northwest, including areas near Washington, Oregon, and British Columbia. This region is known for its complex tectonic interactions, including subduction, where the denser oceanic plate is being forced beneath the lighter continental plate. This area is part of the Cascadia Subduction Zone, which has the potential to produce massive earthquakes. In 1700, a megathrust earthquake estimated to be around magnitude 9.0 struck this area, causing widespread tsunamis and seismic shaking.



Boundary Type:

Reasoning:

Reflect and respond:

1. How did you differentiate between the evidence supporting Wegener's hypothesis of continental drift and the evidence supporting the theory of plate tectonics? Describe the thought process and resources you used to identify these differences.
2. What challenges did you face in understanding the mechanisms proposed by Wegener versus those accepted in the theory of plate tectonics? How did you overcome these challenges?
3. How did learning about the historical context of scientific acceptance impact your understanding of how scientific theories develop and change over time? Provide an example from this activity that illustrates this process.

MODULE 2: MONITOR YOUR PROGRESS

Take a moment to monitor your progress in this module by addressing the questions below:

- Are you using the best strategies to learn this material? If not, what could you do differently?
- Are you going too fast through the module? Are you going too slowly? (One way to reflect on this question is to think about which areas you moved through quickly, and which you took more time on. How is the speed of your progress through the module affecting your learning and the quality of your assessments?)
- Are you understanding the information (or mastering the skills) in this module? If not, what can you do?

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MODULE 2: QUIZ

MODULE 2: QUIZ WRAPPER

- How did you study for this quiz? What strategies did you use to prepare? Which strategies seemed most effective?
- What study practices did not work well? What should you not do, or do differently, next time?
- On which aspects of this quiz did you perform well? Why do you think you performed well on these aspects?
- Can you see patterns in your error and/or specific subjects? How can you address them in future preparation?

MODULE 2: WRAP-UP

THINK ABOUT YOUR LEARNING IN THIS MODULE BY ADDRESSING THE QUESTIONS BELOW:

- How well did you do on this module?
- Which strategies worked best for you for each activity/assignment in this module?
- What are your most significant learnings, or “Aha!” moments, from this module?
- What do these significant learnings, or “Aha!” moments, reveal to you or help you synthesize geology content?

- Before moving on, select the work piece that captures your most significant learning achievement in this module.
- Document and archive this work piece in the [My Core Gallery](#) section of this workbook.